

Examination of PHY Layer Features in DECT-2020 and LTE-M Networks

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Abstract—This paper provides a comparative analysis of the physical layer (PHY) characteristics of DECT-2020 NR and LTE-M, two critical standards for IoT communication. DECT-2020 NR, developed by ETSI, is intended for industrial IoT applications and provides low-latency, reliable communication in high-density environments. In contrast, LTE-M, which is a member of the 3GPP family, is designed to provide broad coverage, energy efficiency, and mobility support for low-power devices. The study emphasizes the distinctions between the standards regarding data rates, modulation schemes, and frequency bands, displaying how each standard fulfills specific IoT requirements. Furthermore, it highlights the complementing characteristics of DECT-2020 NR and LTE-M, by providing performance insights such as bit-error rate (BER) analysis and spectral efficiency.

Index Terms—DECT-2020 NR, LTE-M, IoT

I. INTRODUCTION

The IoT industry has witnessed significant growth and is expected to continue expanding, driven by advancements in data science and artificial intelligence. As new applications emerge, the number of impacted industries is also increasing, each with varying demands. For example, for Industry 4.0, real-time, low latency is a primary requirement, while in others, like environmental monitoring, battery life and energy efficiency are critical [1]. Various IoT standards have been proposed to address these diverse needs over the years.

The 3rd Generation Partnership Project (3GPP) has specified different services to meet these requirements. The major ones are Narrowband IoT (NB-IoT), Long Term Evolution—Machine Type Communication (LTE-M) [2], and New Radio (NR)-based Ultra-Reliable Low-Latency Communications (URLLC). In NB-IoT, the data rate is low, but power consumption is minimal, providing long battery life. It is less expensive than LTE-M owing to its simple technology. LTE-M is similar to NB-IoT but offers higher data rates with mobility support while having a low cost [3]. It is specifically developed for machine-to-machine communication and IoT applications, with support for end-to-end encrypted connections [4]. According to the findings of a recent study in [5],

compared to LTE-M, NB-IoT stands out in scenarios where extensive coverage and deep signal penetration are necessary while maintaining low power consumption. According to [6], NB-IoT improves coverage in attenuated settings like basements and subterranean places. The results show that NB-IoT's single-tone transmission and narrow 200 kHz bandwidth allow it to penetrate challenging locations, such as basements and underground areas, making it particularly well-suited for smart city applications such as water and gas metering or parking sensors installed in subterranean environments. [5]. LTE-M devices can offer maximum of 1 Mbps data rate, while for NB-IoT it is limited to 220 Kbps [7]. Autonomous vehicles and other applications requiring very low latency and high reliability are ideally suited for URLLC, a 5G service.

In response to these needs, the Digital Enhanced Cordless Telecommunications (DECT) working group of the European Telecommunications Standards Institute (ETSI) set up initiatives to create a new IoT communication technology. This technology fits the 5G standards and enables deployment without the need for network infrastructure. Owing to their energy economy, low latency, and high reliability, 5G networks are becoming essential for smart and sustainable city applications [8]. These efforts culminated in 2020 with the introduction of the ETSI-DECT 2020 standard [9].

DECT-2020 New Radio (NR) extends the capabilities of earlier DECT versions, such as DECT and DECT Evolution. It is a 5G Radio Interface Technology standard intended for use in time division duplex (TDD) frequency bands below 6 GHz, in which 17 distinct bands are supported at the physical layer centered at 1.9 GHz carrier frequency [10]. DECT-2020 NR largely employs DECT bands and has elements that allow easy integration with previous DECT systems into its system architecture. The DECT-2020 NR frame and slot lengths (10 ms and 41.6 μ s) and supported channel bandwidth (1.728 MHz - 221 MHz) are identical to the standard DECT. Fortunately, while old DECT was meant to serve

only voice calls, DECT-2020 NR is designed to enable a broader range of services [11]. DECT-2020 NR also utilizes stronger resistance to radio interference, also known as co-channel interference rejection [11]. DECT-2020 NR presents a device-based wireless cluster-tree mesh network that can establish an independent network without additional base stations or a core network. Hybrid low-power wide-area mesh networks have shown mesh-based architectures' potential for IoT applications with their wide-area coverage and energy efficiency [12]. Wireless mesh networking (WMN) allows coverage to function as routers for their neighbors, transferring data between them when a direct connection is not feasible while covering long distances at a minimal cost [13].

Under high device density, the DECT-2020 NR system provides minimal latency, with a 99th percentile delay of no more than 10 ms. Using a multi-hop system with modified bias settings may improve delay performance in DECT-2020 NR. This strategy improves the system's ability to handle applications when low latency is necessary [9]. Owing to its architectural design, which utilizes WMN to decrease interference and allow for spatial reuse of frequency bands, DECT-2020 better enhances system efficiency than LTE-M, particularly in terms of low latency and high node density [14].

In this work, we particularly discuss LTE-M and DECT-2020 NR, focusing on their PHY features and performance across various channel conditions. These two standards have been selected because they address critical IoT communication requirements from different perspectives. LTE-M, part of the 3GPP family, provides an infrastructure-based solution optimized for broad coverage, mobility, and energy efficiency, making it ideal for applications like smart metering, logistics, and asset tracking. On the other hand, DECT-2020 NR, standardized by ETSI, offers a decentralized, infrastructure-free solution, well-suited for high-density, low-latency IoT applications in environments such as smart cities and industrial automation. Rather than a direct comparison, we aim to consider how each of these standards meets the demands of different IoT scenarios, examining their PHY features and analyzing their BER performances.

This paper is organized as follows. In Section II, we introduce the PHY parameters of DECT-2020 NR (1x) and LTE-M (Cat-M1). In Section III, we discuss the achievable data rates for these systems, followed by their BER results in Section IV. Finally, Section V concludes the paper.

II. SYSTEM EVALUATION

Table I compares PHY layer parameters between DECT-2020 NR and LTE-M (Cat-M1), revealing notable differences that reflect their distinct design goals. DECT-2020 NR operates predominantly in the 1.9 GHz fre-

quency band globally, whereas LTE-M utilizes licensed spectrum typically below 1 GHz, such as 700 MHz to 900 MHz, which is advantageous for broader coverage and better penetration through obstacles. In terms of channel bandwidth, DECT-2020 NR supports a wider bandwidth of 1.728 MHz compared to LTE-M's 1.08 MHz, which may impact data rates and spectral efficiency. DECT-2020 NR uses a range of modulation schemes including up to 1024-QAM, whereas LTE-M supports QPSK, 16-QAM, and 64-QAM, enabling higher data rates but requiring more complex receiver designs.

Both systems employ convolutional, turbo, and LDPC codes for channel coding, but their multiplexing methods diverge, with DECT-2020 NR using TDMA and LTE-M leveraging OFDMA and SFDMA in the downlink and uplink respectively. Latency is another distinguishing factor, with DECT-2020 NR achieving sub-10 ms latency, beneficial for applications requiring quick response times, whereas LTE-M typically experiences latency between 10-15 ms. Additionally, while DECT-2020 NR is designed for shorter-range applications with up to a few km, often extended with mesh networks, LTE-M supports a much greater range of 10-15 km, making it more suitable for wide-area deployments. These differences underscore the tailored applications of each technology, with DECT-2020 NR excelling in localized, high-capacity environments and LTE-M offering extensive coverage and higher data rates for broader areas.

Further in Table II, we compare key Orthogonal Frequency Division Multiplexing (OFDM) parameters between DECT-2020 NR and LTE-M downlink, highlighting distinct characteristics. DECT-2020 NR features a subcarrier spacing of 27 kHz, which is wider than LTE-M's 15 kHz, potentially enhancing its performance in reducing interference. Both systems use an FFT size of 64, though it is tailored to their respective subcarrier spacings. DECT-2020 NR employs a cyclic prefix (CP) of 4.63 μ s and a symbol duration of 41.67 μ s, while LTE-M provides options for a normal cyclic prefix of 4.7 μ s and extended options of 16.67 μ s, with corresponding symbol durations of 71.36 μ s and 83.34 μ s. Additionally, DECT-2020 NR supports a variable number of bits per subcarrier (1, 2, 4, 6, 8, or 10), whereas LTE-M is limited to 2, 4, or 6 bits per subcarrier. These differences underscore the design and application-specific optimizations of each OFDM scheme.

III. ACHIEVABLE DATA RATES

We consider the following key factors in calculating the achievable rates of downlink of DECT-2020 and LTE-M: subcarrier spacing Δf , the total number of data subcarriers (N), modulation order (M e.g., 4 for QPSK, 16 for 16-QAM) used in each subcarrier, cyclic prefix

TABLE I
COMPARISON OF PHY LAYER PARAMETERS: DECT-2020 NR VS. LTE-M (REL-13)

Parameter	DECT-2020 NR (1x)	LTE-M (Cat-M1)
Frequency Bands	1.9 GHz (globally)	Licensed spectrum, typically sub-1 GHz (e.g., 700 MHz, 800 MHz, 900 MHz)
Channel Bandwidth	1.728 MHz	1.08 MHz (6 resource blocks of 12x15 KHz)
Modulation	GMSK, $\pi/2$ -BPSK, $\pi/2$ -QPSK, 16-QAM	QPSK, 16-QAM, 64-QAM
Data Rate	Up to several Mbps	Up to 1 Mbps (1.4 MHz bandwidth)
Duplexing Method	TDD (Time Division Duplex)	FDD (Frequency Division Duplex) and TDD
Maximum Transmit Power	23 dBm	20 dBm
Receiver Sensitivity	-102 dBm	-132 dBm to -140 dBm
Spectral Efficiency	1 to 4 bits/s/Hz	Up to 2.5 bits/s/Hz
Coding Schemes	Convolutional, Turbo, and LDPC codes	Convolutional, Turbo, and LDPC codes
Multiplexing	TDMA (Time Division Multiple Access)	OFDMA (Orthogonal Frequency Division Multiple Access), SFDMA (Single Carrier Frequency Division Multiple Access)
Channel Coding	Convolutional Coding (with optional Turbo or LDPC)	Turbo Coding (with optional LDPC)
Latency	Sub-10 ms (low latency)	10-15 ms
MIMO Support	Optional	Optional (Single-Input Single-Output or 2x2 MIMO)
Synchronization	TDMA-based frame synchronization	OFDMA-based frame synchronization
Error Detection	CRC (Cyclic Redundancy Check)	CRC (Cyclic Redundancy Check)
Handover Support	Not applicable (for mesh networks)	Supported (within LTE network)
Maximum Range	Up to 300 meters (extendable with mesh)	Up to 10-15 km (depending on deployment)

TABLE II
COMPARISON OF OFDM PARAMETERS: DECT-2020 NR VS. LTE-M

OFDM Parameter	DECT-2020 NR (1x)	LTE-M (Cat-M1)
Subcarrier Spacing (Δf)	27 kHz	15 kHz (standard LTE)
FFT Size	64 (for 27 kHz subcarrier spacing)	64 (for 15 kHz subcarrier spacing)
Cyclic Prefix Duration (T_{cp})	4.63 μ s (8 unit)	4.7 μ s (Normal CP), 16.67 μ s (Extended CP)
Symbol Duration (T_{sym})	41.67 μ s (72 unit)	71.36 μ s (normal CP), 83.34 μ s (extended CP)
Number of data subcarriers per OFDM symbol	52 unit	10 of 12 unit per OFDM symbol (the exact number may vary slightly depending on the specific configuration).
Number of bits per subcarrier	1, 2, 4, 6, 8 or 10	2, 4, 6

duration and OFDM symbol duration including cyclic prefix (T_{sym}). The capacity C (bps) can be calculated using

$$C = N \times \log_2(M) \times \left(\frac{1}{T_{sym}} \right). \quad (1)$$

The OFDM symbol duration can be further represented as $T_{sym} = T_{fft} + T_{cp}$ where $T_{fft} = \frac{1}{\Delta f}$ is the duration of the FFT (Fast Fourier Transform) process, which is the inverse of the subcarrier spacing Δf . T_{cp} is the cyclic prefix duration, often expressed as a fraction of T_{fft} .

- **LTE-M¹** : Achievable capacity with 64-QAM is approximately 4.31 Mbps and 5.04 Mbps respectively for extended and normal CP settings. Note that, extending CP provides extra protection against intersymbol interference due to larger delay spread, however, it reduces the achievable rates.

¹These calculations are based on the versions in Table I and can vary depending on specific configurations.

- **DECT-2020 NR¹** : Achievable capacity is approximately 12.47 Mbps with 1024-QAM for the example parameters provided in Table II.

Fig. 1 shows achievable rates versus number of bits per subcarrier for the two standards. DECT-2020 NR generally provides higher capacity due to higher subcarrier count and higher modulation order within shorter OFDM symbol duration. It is further optimized for low-latency and reliable communication in industrial environments, which might not always prioritize maximum throughput but focuses on robustness and energy efficiency. LTE-M, on the other hand, is designed primarily for broader coverage and extended battery life, supporting a variety of low-power devices over wide geographic areas.

IV. RELIABILITY RESULTS FOR DECT-2020 AND LTE-M

To assess the reliability performance of the two standards, we run bit-error-rate (BER) simulations for the parameters given in Tables I and II. In our analysis, the path loss (PL) is calculated using the COST 231

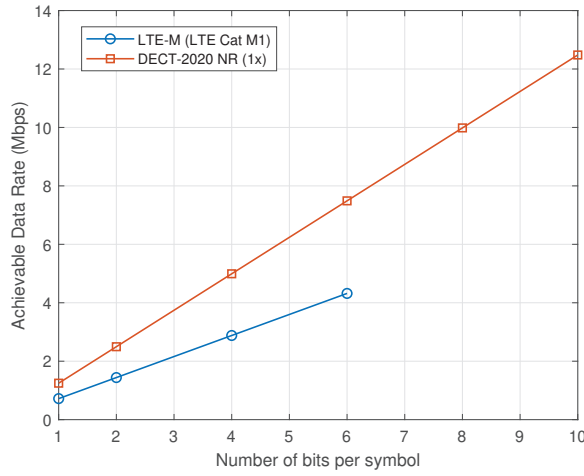


Fig. 1. Data Rate vs Modulation Order for LTE-M (LTE Cat M-1) and DECT-2020 (1x) and the parameters in Table II.

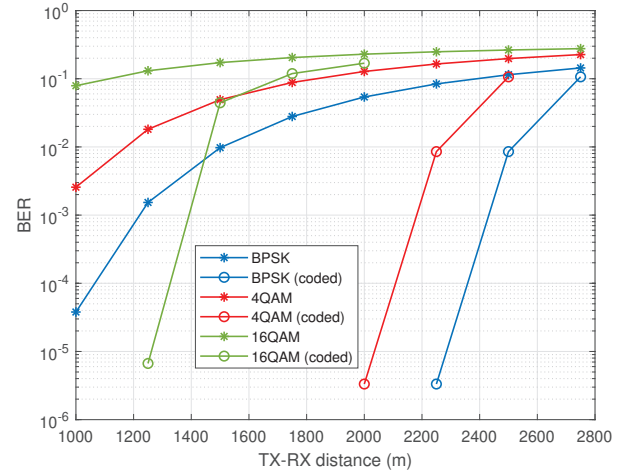


Fig. 2. BER of DECT-2020 (1x) versus TX-RX distance for the specifications in Tables I and II.

Walfisch Ikegami model, which was proposed for frequencies in the range of 800 to 2000 MHz, as $PL = 42.6 + 26 \log_{10}(d_m) + 20 \log_{10}(f)$ where d_m is the distance between transmitter and receiver, and f is the carrier frequency [15]. The operating frequencies for DECT-2020 NR and LTE-M are 1.9 GHz and 700 MHz, respectively, with transmitter powers set at 20 dBm. The noise spectral density is -174 dBm/Hz and the noise figure is taken is 8 dB in both analyses. The channel is assumed to be flat and only additive white Gaussian noise is considered at the receiver.

In Fig. 2, we present the BER results versus the transmitter-receiver distance for DECT-2020 for modulation orders 2, 4, and 16 with and without coding. Turbo coding with a rate of 1/3 has been implemented. While coded implementation provides better performance, for a target BER of 10^{-3} , the coverage with BPSK can go up to 2.4km. For larger modulation orders, e.g., 16-QAM on the other hand, the coverage shrinks down to around 1.4km. In Fig. 3, we present the BER results for LTE-M; this time coverage extends significantly as compared to DECT-2020 NR with the cost of reduced data rates as discussed in the previous section. The increase in coverage results from the reduced noise bandwidth (from 27 KHz to 15 KHz per subcarrier) and lower carrier frequency, which lead to less attenuation during propagation, and overall higher signal-to-noise ratio at the receiver. For the same target BER, LTE-M can cover a region up to 7-8 km with coded BPSK. For larger modulation orders, with coded 16-QAM, the coverage is around 3-4 km.

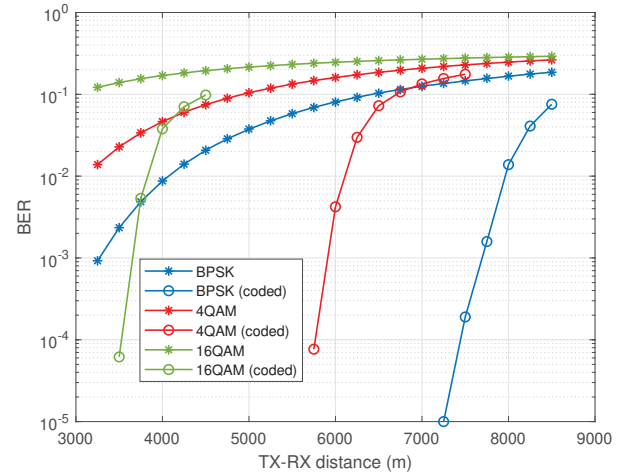


Fig. 3. BER of LTE-M (LTE Cat M-1) versus TX-RX distance for the specifications in Tables I and II.

V. CONCLUSION

In this study, we discussed the physical layer features of two popular IoT standards, LTE-M and DECT-2020, and presented the results of their achievable rates and BER. Our work, however, primarily focused on numerical simulations under certain assumptions and specific versions of the two standards. While our analyses provide valuable insights, they represent only a fraction of the broader picture. In practice, evaluating the effectiveness of these standards requires a comprehensive consideration of many factors, including energy efficiency, latency, reliability, coverage, deployment cost, spectrum availability, and the specific application scenarios for which they are intended. The diverse use cases for LTE-

M and DECT-2020 NR, such as LTE-M's focus on wide-area IoT networks and DECT-2020 NR's emphasis on industrial IoT and private networks, further challenge the direct comparisons. These standards are optimized for different environments and requirements, making it difficult to declare one as superior to the other.

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